



END-TO-END EXPLORATORY DATA ANALYSIS OF A PUBLIC DATASET USING PYTHON



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1 INTRODUCTION

Exploratory Data Analysis (EDA) is the process of inspecting, cleaning, summarising and visualising data before formal modelling. A complete Python-based EDA workflow helps transform a public dataset from raw records into reliable evidence by checking data quality, understanding distributions, finding relationships and communicating the most important patterns.

DATASET IRIS 150 labelled observations	FEATURES 4 NUMERIC sepal + petal measures	TARGET 3 CLASSES flower species
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PROBLEM STATEMENT

- Raw public datasets may contain missing values, inconsistent types and patterns that are not immediately visible.
- Without systematic EDA, important distributions and feature relationships can be overlooked.

OBJECTIVES

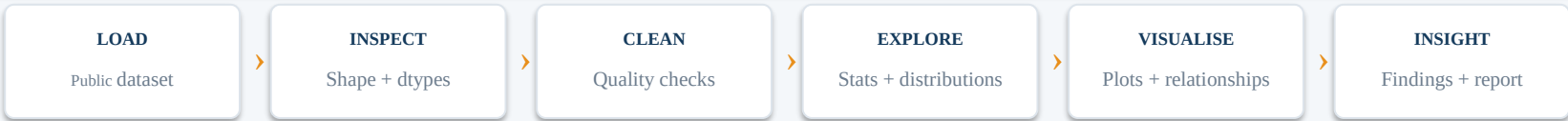
- Load and inspect a public dataset using Python.
- Clean and validate data quality before analysis.
- Use descriptive statistics and visualisations to identify patterns.
- Summarise insights in a reproducible end-to-end workflow.

2 DATASET & STUDY DETAILS

DATASET IRIS public benchmark	ROWS 150 observations	FEATURES 4 numeric variables	CLASSES 3 species labels	TOOLS PYTHON EDA stack
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- The Iris dataset contains sepal length, sepal width, petal length and petal width for three flower species.
- Each species contributes 50 observations, making the target classes balanced in the standard dataset.
- Python tools such as pandas, NumPy, Matplotlib and Seaborn can support loading, validation, descriptive statistics and visualisation.

3 ARCHITECTURE / METHODOLOGY



- The workflow starts with data ingestion and structure checks, followed by missing-value and duplicate inspection. Descriptive statistics then explain central tendency and spread; visualisations reveal distributions, class balance and feature relationships. The final stage converts the evidence into concise, reproducible insights.

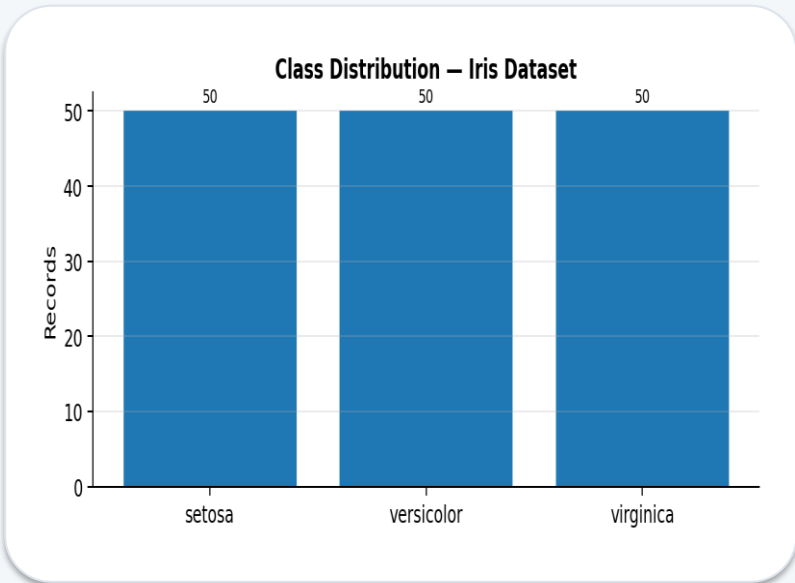
4 KEY EDA FINDINGS

DATA QUALITY 150 / 150 Complete standard Iris dataset; no missing values in the classic version.	CLASS BALANCE 50 EACH Each of the three species has the same number of records.	FEATURE SEPARATION PETAL > SEPAL Petal length and petal width provide stronger visual separation between species.	RELATIONSHIPS POSITIVE Petal measurements show clear positive association, especially within species groups.
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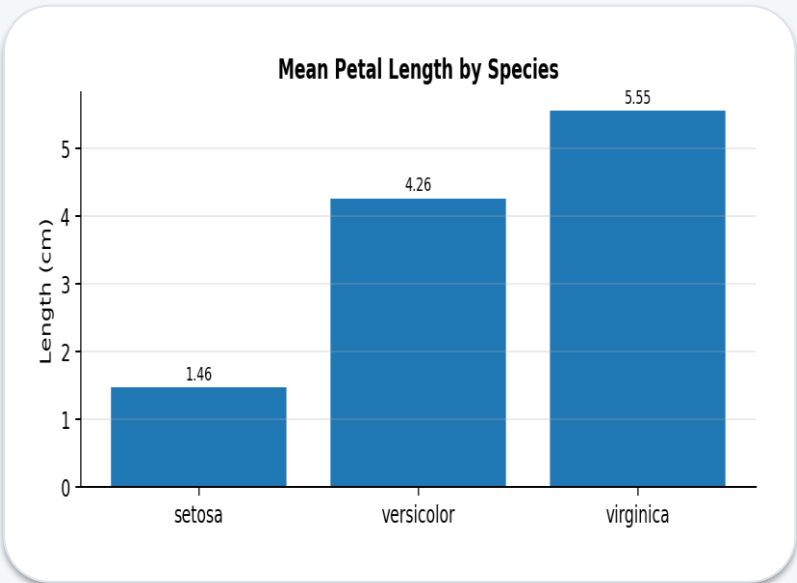
5 RESULTS — PYTHON EDA

RECORDS 150 Iris observations	FEATURES 4 numeric measurements	CLASSES 3 balanced species	MISSING 0 standard dataset	BEST SEPARATOR PETAL length / width
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CLASS DISTRIBUTION



MEAN PETAL LENGTH



STATISTICAL RESULTS

- Mean petal length: 3.76 cm
- Mean petal width: 1.20 cm
- All three classes contain 50 records
- No missing values in the standard dataset

KEY OBSERVATIONS

Petal measurements provide clearer class separation than sepal measurements. The balanced class structure makes the dataset useful for demonstrating a complete EDA workflow before modelling.

6 CONCLUSION & RECOMMENDATIONS

CONCLUSION

The end-to-end Python EDA workflow demonstrates how a public dataset can be systematically transformed into useful evidence. Starting from ingestion and data-quality checks, the analysis progresses through descriptive statistics and visualisation to identify the strongest patterns. In the Iris dataset, class balance is clear and petal measurements provide the most useful visual separation among species.

RECOMMENDATIONS

- Reproducible workflow**
Keep cleaning, analysis and visualisation steps documented in Python.
- Visual evidence**
Use multiple plots to validate numerical findings and avoid misleading conclusions.

- Quality first**
Check missing values, duplicates, types and outliers before interpreting patterns.
- Extend the EDA**
Add correlation analysis, pair plots and subgroup comparisons before modelling.